

AI Customer Support Ticket Classifier

Technical Assignment Report

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1. Project Overview

This project implements a basic machine learning system for automatically classifying customer support tickets based on their descriptions. The system converts ticket text into numerical features using TF-IDF and uses Logistic Regression to predict the ticket category. A Streamlit web application is also provided so that a user can enter a new ticket and receive a predicted category together with a confidence score.

2. Problem Statement

Customer support teams receive many requests through different channels. Manually sorting these requests into categories can take time. The goal of this project is to build a simple classifier that can automatically identify the category of a support ticket from its text description.

3. Objectives

- Create and analyze a customer support ticket dataset.
- Preprocess ticket descriptions for machine learning.
- Convert text into numerical features using TF-IDF.
- Train a Logistic Regression classification model.
- Evaluate the model using Accuracy, Precision, Recall and F1 Score.
- Test the model on new ticket descriptions.
- Provide a simple Streamlit interface for interactive prediction.

4. Dataset

A custom dataset containing 225 customer support ticket records was prepared for this assignment. There are five categories, with 45 records in each category.

Field	Description
ticket_id	Unique ticket identifier
ticket_description	Description of the customer's issue
category	Target category to predict
priority	Priority of the ticket
status	Current ticket status

Categories: Login Issue, Application Error, Report, Account Update, Performance.

5. Exploratory Data Analysis

Basic exploratory analysis was performed to understand the dataset. The analysis included category distribution, priority distribution, status distribution and general dataset checks. Category and priority charts were generated and stored in the outputs directory.

6. Data Preprocessing

The preprocessing stage loads the CSV dataset, checks the data, prepares the ticket descriptions, and converts the text to lowercase before it is passed to the TF-IDF vectorizer. The preprocessing code is available in `src/preprocessing.py`.

7. Machine Learning Approach

7.1 TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency) is used to represent ticket descriptions as numerical feature vectors. The approach gives useful words more importance while reducing the effect of words that occur frequently across documents.

7.2 Logistic Regression

Logistic Regression was selected as the classification algorithm because it is simple, fast, and commonly used for text classification. The trained model is also able to provide class probabilities, which are used as the displayed confidence score.

7.3 Train-Test Split

The dataset was divided into 80% training data and 20% testing data. The model was trained using the training portion and evaluated on the held-out test portion.

8. Model Evaluation

Metric	Score
Accuracy	100.00%
Precision	100.00%
Recall	100.00%
F1 Score	100.00%

The current test split produced a perfect score across the four reported metrics. This result should be interpreted carefully because the dataset is small, custom-created for the assignment, and contains relatively distinct category patterns. It should not be treated as production-level validation.

A confusion matrix was generated during training and is available as `outputs/confusion_matrix.png`.

9. New Ticket Testing

#	Test Description	Prediction	Confidence
1	Forgot password and cannot login	Login Issue	77.52%
2	Application crashes when saving	Application Error	70.91%
3	Need monthly sales report	Report	73.36%
4	Change registered mobile number	Account Update	69.70%
5	Website is taking too long to load	Performance	51.04%

The five new ticket descriptions were classified correctly according to the intended categories. The Performance example had the lowest confidence at 51.04%, showing that a correct prediction can still have relatively low confidence

when the wording is less distinctive.

10. Streamlit Application

A simple Streamlit interface was created to make the classifier easier to use. The user can enter a ticket description and click the Predict Ticket button. The application displays the predicted category and confidence score. The application is launched with:

```
python -m streamlit run src/app.py
```

11. Project Structure

```
AI_Assignment_Suvadip/  
├── data/  
│   └── tickets.csv  
├── src/  
│   ├── create_dataset.py  
│   ├── preprocessing.py  
│   ├── train.py  
│   ├── predict.py  
│   └── app.py  
├── model/  
│   ├── ticket_classifier.pkl  
│   └── tfidf_vectorizer.pkl  
├── outputs/  
│   ├── category_distribution.png  
│   ├── priority_distribution.png  
│   └── confusion_matrix.png  
├── screenshots/  
│   ├── dataset.png  
│   ├── analysis.png  
│   ├── confusion_matrix.png  
│   └── prediction.png  
├── requirements.txt  
└── README.md
```

12. How to Run

```
python -m pip install -r requirements.txt
```

```
python src/create_dataset.py
```

```
python src/preprocessing.py
```

```
python src/train.py
```

```
python src/predict.py
```

```
python -m streamlit run src/app.py
```

13. Limitations

- The dataset is relatively small.
- Only five categories are included.
- The dataset was created for the assignment rather than collected from a production support system.
- Short or ambiguous ticket descriptions may receive lower confidence.
- The model has not been evaluated on a large real-world dataset.

14. Future Improvements

- Use a larger and more diverse real-world ticket dataset.
- Add additional support categories.
- Compare multiple machine learning algorithms.
- Use cross-validation and hyperparameter tuning.
- Store tickets and predictions in a database.
- Add an API for integration with other applications.
- Use corrected predictions as feedback for future model training.

15. AI Assistance Disclosure

AI tools were used during development to help understand the assignment requirements, suggest project structure and implementation ideas, debug development/environment issues, and improve documentation. The implementation was reviewed and tested locally before submission.

16. Conclusion

The project demonstrates an end-to-end machine learning workflow for customer support ticket classification. It covers dataset preparation, preprocessing, exploratory analysis, TF-IDF feature extraction, Logistic Regression training, model evaluation, new-ticket prediction, and a Streamlit-based user interface. The current implementation provides a working foundation that can be extended with a larger real-world dataset and additional production features.

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